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An essential message is woven throughout the first three parts of this series on Design Thinking – but it is subtle and easily missed [1-3]. This article faces it head on. It is the concept of Framing and Reframing the design problem.

In short, problem framing is about uncovering the actual problem worth solving – which is often hidden to everyone when the design process begins. Framing and reframing are essential elements of design because "it is often not at all clear what 'the problem' is, it may have been only loosely defined by the client, many constraints and criteria may be un-defined, and everyone involved in the project may know that goals may be re-defined during the project" [5].

[1] C. Mattson, "Design Thinking Part 4: Framing and Reframing Design Problems," *The BYU Design Review*, Aug. 14, 2023. [Online]. Available: https://www.designreview.byu.edu/collections/design-thinking-part-4-framing-and-reframing-design-problems. [Accessed: Apr. 9, 2026].

What is brainwriting?

Also known as 6-3-5 brainwriting or the 6-3-5 method, brainwriting is a practice designed to stimulate creativity and help groups generate ideas. Originally developed in the late 1960s by Bernd Rohrbach, a German marketer, it guides a small group of people through several rounds of communal ideation and refinement.

Six people identify a problem statement, i.e., what they're trying to solve. Each participant then writes this statement at the top of a rubric (known as the Idea Form). Then, they try to generate three ideas in five minutes. Six people, three ideas each, in five minutes—or 6-3-5.

Unlike traditional verbal brainstorming, the ideas are written out or expressed

[2] H. Herman, "What is brainwriting? The 6-3-5 method explained (with template)," *Zapier*, Nov. 14, 2024. [Online]. Available: https://zapier.com/blog/brainwriting/. [Accessed: Apr. 8, 2026].

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The benefits of iterative prototyping

The benefits of iterative prototyping cannot be understated. Iterative prototyping allows for quick feedback and iteration, helps to identify and fix bugs early on, improves user experience, and can lead to cost savings.

In this section, we will explore each of these benefits in more detail and explain why they make iterative prototyping an essential aspect of application development.

Allows for quick feedback and iteration

Iterative prototyping allows for quick feedback and iteration by breaking down the development process into smaller stages. This allows for early identification of user needs and faster development time.

By testing a prototype with users, developers can gather valuable feedback and make adjustments to the design, features, and functionality of the app. This helps to ensure that the final product meets the needs of the target audience and is more likely to be successful in the marketplace.

[3] C. Fanchi, "The Importance of Iterative Prototyping in Application Development," *Backendless*, Feb. 7, 2023. [Online]. Available: https://backendless.com/the-importance-of-iterative-prototyping-in-application-development/. [Accessed: Apr. 8, 2026].

Purpose of CAD

Used by engineers, architects, and construction managers, CAD has replaced manual drafting in many places. It helps users create designs both 2D and 3D designs to better visualize construction.

CAD enables the development, modification, and optimization of the design process. Engineers can make more accurate representations and modify them easily to improve design quality. The software also takes into account how various materials interact: This is especially relevant as more details are added to drawings by subcontractors.

[4] L. Bernstein, "Computer-Aided Design (CAD): What It Is — and Why It's Important," *Procore*, Sep. 23, 2025. [Online]. Available: https://www.procore.com/library/computer-aided-design. [Accessed: Apr. 11, 2026].

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What is pairwise comparison?

The pairwise comparison method (sometimes called the *paired comparison method*) is a process for ranking or choosing from a group of alternatives by comparing them against each other in pairs, i.e. two alternatives at a time. Pairwise comparisons are widely used for decision-making, voting and studying people's preferences.

How does the pairwise comparison method work?

The pairwise comparison method works by each alternative being compared against every other alternative in pairs – i.e. "head-to-head". The decision-maker pairwise *ranks* the alternatives in each pair: by deciding which one is higher ranked or if they are equally ranked.

[5] 1000minds, "Pairwise comparison method & pairwise ranking," *1000minds*, 2026. [Online]. Available: https://www.1000minds.com/decision-making/pairwise-comparison. [Accessed: Apr. 10, 2026].

Design verification and validation are integral to the digital product development process. The verification procedure ensures the project is built according to the agreed requirements, while validation analyzes the result to meet users' needs. These terms go hand in hand and are usually confused with one another. Design verification and validation in product lifecycle should always be completed. Verification directly influences product quality and performance. Validation determines the user's successful adoption of the project.

Neglecting those processes doesn't only lead to creating technically poor products that evoke customer dissatisfaction. However, this also violates the key requirements of ISO 9001:2015 — a globally acknowledged benchmark for quality management. To avoid this scenario, find an explanation of design verification and validation.

WHAT IS DESIGN VALIDATION?

Design validation is the procedure of estimating the product's capability to meet user needs and operate in a real-world environment. It is conducted after development to address the question: "Did we create the right product"? Design validation aims to build a bridge between end customers, including stakeholders' expectations and engineering needs. Even if your project meets all the technical requirements, it does not guarantee

[6] G. Shchebakov, "What Is Design Validation and Verification and Why It's Crucial," *White Test Lab*, Dec. 4, 2024. [Online]. Available: https://white-test.com/for-qa/useful-articles-for-qa/design-verification-validation/. [Accessed: Apr. 11, 2026].

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Functional Decomposition

Functional decomposition is a technique for dividing a larger task into smaller tasks. During decomposition, the team focuses on the inputs and outputs of each task and uses generic descriptions of the process achieved by the task. The decomposition process is usually applied recursively until the design team can readily envision ways of achieving each of the sub-tasks. Functional decomposition is also called functional analysis, top-down design, or logical decomposition.

Functional decomposition can be applied to many engineering tasks. For *our purposes*, i.e. in ME 491-493, we use functional decomposition as a tool in the conceptual design phase. During conceptual design, the team uses functional decomposition to explore solutions to a given problem without overly constraining the choices for implementing those solutions. That is consistent with the "conceptual" part of Concept Design. Implementation details are workout out in the Subsystem Engineering phase (to use the terminology of Mattson and Sorenson).

A functional decomposition for a design goal is linked to a specific design concept. One should not expect one-and-only-one functional decomposition for a given objective. Since the design team will need to choose among competing conceptual designs, the team should expect to develop more than one functional decomposition. For example, consider the task of making coffee. Different functional decompositions would be needed to describe the a drip coffee maker, an espresso machine, or a machine that dispenses coffee made from "instant" powdered coffee.

Also remember that design will require iteration and refinement as the team's knowledge improves as the design evolves. Therefore, initial attempts at functional decomposition will likely need to be revised or even discarded altogether. The goal is greater understanding, not checking a box in some recipe for the design process.

[7] G. Recktenwald, "Functional Decomposition," *Portland State University*, 2026. [Online]. Available: http://web.cecs.pdx.edu/~gerry/class/ME491/notes/functional_decomposition.html. [Accessed: Apr. 1, 2026].

The Takeaway

Stakeholder interview is an important aspect of UX design process. It is the foundation of an entire project. The process provides not only the quantitative information but the board overview of the project as well. Engaging stakeholders throughout the project life cycle is key to a project's success, however not a surety. Managing stakeholder expectations and ensuring their active involvement is very much important to a project for the continuation of a project and its successful completion.

Whether you're working with a large enterprise, an up-and-coming startup, or a small family-owned business, start planning your stakeholder interviews can ensure creation of designs that are user friendly, practical, and meet the business goals.

[8] Evoke Technologies, "Importance of Stakeholders Analysis in Design Thinking," *Evoke Technologies*, Mar. 28, 2017. [Online]. Available: https://www.evoketechnologies.com/blog/uncategorized/importance-of-stakeholders-analysis-in-design-thinking/. [Accessed: Apr. 11, 2026].

[9] University of Toronto, "ESC101 Design Report - Soggy Sandwiches" University of Toronto, Design Report, 2025. [Online]. Available: https://docs.google.com/document/d/1nZOmCTX_TPwlIUAHHwPeHPWsxBL09zqfqFC4CfeZKIs/edit?tab=t.0 [Accessed: Apr. 7, 2026].

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[10] University of Toronto, “CIV102 Project Team 501 Design Report” University of Toronto, Bridge Project, 2025. [Online]. Available: https://docs.google.com/document/d/13Ov8X5e0ZJSeNW6gkUP55DKxJJ5aCkA47Wyev7CGkkI/edit?tab=t.0 [Accessed: Apr. 7, 2026].

[11] RFP_F_-_Accessible_Filming Discussions,” Quercus, University of Toronto, 2026. [Online]. Available: Quercus Course Page. [Accessed: Mar. 31, 2026].